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Inventor(s)	Mutnury; Bhyrav et al.

System for crosstalk rejecting topology

Abstract

A system for preventing crosstalk between adjacent channels comprises a crossover connector positioned along a length of one channel such that a portion of a positive trace for a first channel is positioned adjacent to a positive trace of a positive trace of an adjacent channel. The position of the crossover connector is based on preventing crosstalk and crossover connectors on adjacent channels may be staggered to further prevent crosstalk. A crossover connector may be based on capacitors or resistors to prevent crosstalk.

Inventors: Mutnury; Bhyrav (Round Rock, TX), Farkas; Sandor (Round Rock, TX)

Applicant: Dell Products L.P. (Round Rock, TX)

Family ID: 1000008781113

Assignee: Dell Products L.P. (Round Rock, TX)

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Primary Examiner: Baltzell; Andrea Lindgren

Assistant Examiner: Glenn; Kimberly E

Attorney, Agent or Firm: McDermott Will & Schulte LLP

Background/Summary

BACKGROUND

Field of the Disclosure

(1) This disclosure relates generally to information handling systems and, more particularly, to systems for crosstalk rejecting communication.

Description of the Related Art

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(3) High performance servers implement high speed channels using differential pair routing. As processing and communication speeds increase, crosstalk along long adjacent traces is becoming more common.

SUMMARY

(4) Embodiments disclosed herein may be generally directed to systems for preventing crosstalk between adjacent channels in high performance information handling systems.

- (5) Embodiments may be generally directed to a system comprising a first channel formed with a first positive trace having a first positive trace length, a first negative trace having a first negative trace length, and a capacitor positioned at a first location along the first positive trace length, wherein the first positive trace is on a first side of the first negative trace. The system further comprises a second channel having a second channel length. The second channel comprises a second positive trace comprising a first positive trace portion having a first positive trace portion length and a second positive trace portion having a second positive trace portion length, a second negative trace comprising a first negative trace portion having a first negative trace portion length and a second negative trace portion having a second negative trace portion length, and a crossover connector comprising a positive trace crossover capacitor or resistor and a negative trace crossover capacitor or resistor.
- (6) The first positive trace portion of the second positive trace is on the first side of the first negative trace portion of the second negative trace and is coupled to a first positive trace post of the crossover connector, wherein the first negative trace portion of the second negative trace is coupled to a first negative trace post of the crossover connector. The second positive trace portion of the second positive trace is on the second side of the second negative trace portion of the second negative trace and is coupled to a second positive trace post in the crossover connector, wherein the second negative trace portion of the second negative trace is coupled to a second negative trace post.
- (7) In some embodiments, the crossover connector is positioned a distance from a signal source such as a processor. In some embodiments, the crossover connector is positioned a distance less than half the length of the first positive trace length. In some embodiments, the crossover connector is located on a motherboard. In some embodiments, the crossover connector is located on a card.
- (8) In some embodiments, the crossover connector comprises a plurality of layers, wherein each layer comprises two positive post pads and two negative post pads, wherein at least one layer comprises a filament between the two positive pads, at least one layer comprises a filament between the two negative pads and at least one layer comprises does not have a filament between the two positive pads or the two negative pads.
- (9) In some embodiments, the crossover connector comprises a resistor. In some embodiments, the crossover connector comprises a capacitor.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the invention and its features and advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a block diagram of an information handling system;
- (3) FIG. 2 depicts a layout of a plurality of pairs of traces, illustrating a situation in which crosstalk can occur between adjacent traces;
- (4) FIG. 3 depicts a layout of a plurality of channels according to one embodiment, illustrating a system for preventing crosstalk between adjacent traces;
- (5) FIG. 4 depicts a layout of a plurality of channels with all caps near a source, illustrating a situation in which crosstalk can occur between adjacent traces;
- (6) FIG. 5 depicts a layout of a plurality of channels according to one embodiment, illustrating a system for preventing crosstalk between adjacent traces using capacitors;
- (7) FIG. 6 depicts a layout of a plurality of channels according to one embodiment, illustrating a system for preventing crosstalk between adjacent traces using a capacitor crossover connector;
- (8) FIG. 7 depicts a layout of a plurality of channels according to one embodiment, illustrating a

system for preventing crosstalk between adjacent traces using a resistor crossover connector;

(9) FIG. **8** is a top view of layers of a crossover connector package according to one embodiment of a system for preventing crosstalk between adjacent traces;

(10) FIG. **9A** is a perspective exploded view of layers of a crossover connector package according to one embodiment of a method for manufacturing a system for preventing crosstalk between adjacent traces;

(11) FIG. **9B** is a perspective view of an assembled crossover connector package according to one embodiment of a method for manufacturing a system for preventing crosstalk between adjacent traces;

(12) FIG. **10** is a close-up partial perspective view of layers of the crossover connector package of FIG. **9B** according to one embodiment of a method for manufacturing a system for preventing crosstalk between adjacent traces;

(13) FIG. **11** is a perspective view of an assembled capacitor package with metallization applied to the layers according to one embodiment of a method for manufacturing a system for preventing crosstalk between adjacent traces; and

(14) FIG. **12** is a block diagram illustrating selective placement of a system for preventing crosstalk in an information handling system.

DESCRIPTION OF PARTICULAR EMBODIMENT(S)

(15) In the following description, details are set forth by way of example to facilitate discussion of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed embodiments are exemplary and not exhaustive of all possible embodiments.

(16) For the purposes of this disclosure, an information handling system may include an instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize various forms of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, an information handling system may be a personal computer, a consumer electronic device, a network storage device, or another suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include memory, one or more processing resources such as a central processing unit (CPU) or hardware or software control logic. Additional components of the information handling system may include one or more storage devices, one or more communications ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and one or more video displays. The information handling system may also include one or more buses operable to transmit communication between the various hardware components.

(17) Turning to the drawings, FIG. **1** illustrates a block diagram depicting selected elements of an embodiment of information handling system **100**. It is noted that FIG. **1** is not drawn to scale but is a schematic illustration.

(18) As shown in FIG. **1**, components of information handling system **100** may include, but are not limited to, a processor subsystem **12**, which may comprise one or more processors, and a system bus **14** that communicatively couples various system components to processor subsystem **12** including, for example, a memory subsystem **16**, an I/O subsystem **18**, local storage resource **20**, and network interface **22**.

(19) Processor subsystem **12** may comprise a system, device, or apparatus operable to interpret and execute program instructions and process data, and may include a microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), or another digital or analog circuitry configured to interpret and execute program instructions and process data. In some embodiments, processor subsystem **12** may interpret and execute program instructions and process data stored locally (e.g., in memory subsystem **16**). In the same or alternative embodiments, processor subsystem **12** may interpret and execute program instructions and process data stored

remotely (e.g., in a network storage resource).

(20) System bus **14** may refer to a variety of suitable types of bus structures, e.g., a memory bus, a peripheral bus, or a local bus using various bus architectures in selected embodiments. For example, such architectures may include, but are not limited to, Micro Channel Architecture (MCA) bus, Industry Standard Architecture (ISA) bus, Enhanced ISA (EISA) bus, Peripheral Component Interconnect (PCI) bus, PCI-Express bus, HyperTransport (HT) bus, and Video Electronics Standards Association (VESA) local bus.

(21) Memory subsystem **16** may comprise a system, device, or apparatus operable to retain and retrieve program instructions and data for a period of time (e.g., computer-readable media).

Memory subsystem **16** may comprise random access memory (RAM), electrically erasable programmable read-only memory (EEPROM), a PCMCIA card, flash memory, magnetic storage, opto-magnetic storage, and/or a suitable selection and/or array of volatile or non-volatile memory that retains data after power to its associated information handling system, such as system **100**, is powered down.

(22) In information handling system **100**, I/O subsystem **18** may comprise a system, device, or apparatus generally operable to receive and transmit data to or from or within information handling system **100**. I/O subsystem **18** may represent, for example, a variety of communication interfaces, graphics interfaces, video interfaces, user input interfaces, and peripheral interfaces. In various embodiments, I/O subsystem **18** may be used to support various peripheral devices, such as a touch panel, a display adapter, a keyboard, a touch pad, or a camera, among other examples. In some implementations, I/O subsystem **18** may support so-called 'plug and play' connectivity to external devices, in which the external devices may be added or removed while information handling system **100** is operating.

(23) Local storage resource **20** may comprise computer-readable media (e.g., hard disk drive, floppy disk drive, CD-ROM, and other type of rotating storage media, flash memory, EEPROM, or another type of solid-state storage media) and may be generally operable to store instructions and data.

(24) Network interface **22** may be a suitable system, apparatus, or device operable to serve as an interface between information handling system **100** and a network (not shown). Network interface **22** may enable information handling system **100** to communicate over a network using a suitable transmission protocol or standard. In some embodiments, network interface **22** may be communicatively coupled via a network to a network storage resource (not shown). A network coupled to network interface **22** may be implemented as, or may be a part of, a storage area network (SAN), personal area network (PAN), local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a wireless local area network (WLAN), a virtual private network (VPN), an intranet, the Internet or another appropriate architecture or system that facilitates the communication of signals, data and messages (generally referred to as data). A network coupled to network interface **22** may transmit data using a desired storage or communication protocol, including, but not limited to, Fibre Channel, Frame Relay, Asynchronous Transfer Mode (ATM), Internet protocol (IP), other packet-based protocol, small computer system interface (SCSI), Internet SCSI (iSCSI), Serial Attached SCSI (SAS) or another transport that operates with the SCSI protocol, advanced technology attachment (ATA), serial ATA (SATA), advanced technology attachment packet interface (ATAPI), serial storage architecture (SSA), integrated drive electronics (IDE), or any combination thereof. A network coupled to network interface **22** or various components associated therewith may be implemented using hardware, software, or any combination thereof.

(25) Components described above may communicate with each other over channels, wherein each channel may comprise a pair of traces. One major challenge for information handling systems is lowering signal losses associated with the traces. For example, high performance servers implement high speed channels using differential pair routing. As speeds increase, the pair-to-pair

isolation needs to increase to prevent crosstalk along long adjacent traces. However, any extra spacing reduces routing density and increases layer count and cost.

(26) Embodiments disclosed herein may comprise pairs of traces with alternating polarity over some portion of their length to reduce crosstalk. Alternating polarity may be accomplished by staggering connectors of adjacent channels or implementing crossover connectors.

(27) Referring to FIG. 2, information handling systems, such as high performance servers, implement high speed channels **10** using differential pair routing. As depicted in FIG. 2, a first channel **110-1** comprises a differential pair including positive trace **112** and negative trace **114**. Each trace **112**, **114** comprises connector **116** near the source (not shown). Each channel **110** (e.g., channel **110-1**) is separated from an adjacent channel **110** (e.g., channel **110-2**) by a distance **118-1**. For information handling systems **100** operating at lower power or speeds, distance **118** may be small. However, as speeds increase, the pair-to-pair isolation needs to increase to prevent crosstalk along long adjacent traces. However, increasing the distance **118** between adjacent channels **110** may not be possible, since extra spacing reduces routing density and increases layer count and cost.

(28) Embodiments disclosed herein may configure the traces of a channel to cross over each other along a length of the channel **110**, whereby signals from adjacent channels **110** do not cross over and signal losses are reduced.

(29) Particular embodiments are best understood by reference to FIGS. 3-8, 9A-9B and 10-11, wherein like numbers are used to indicate like and corresponding parts.

(30) Referring to FIG. 3, a system for preventing crosstalk between channels **310** having positive traces **112** and negative traces **114** may comprise a staggered connector configuration, wherein a first channel **310-1** has first connectors **116** at a first position between the ends of the first channel **310-1** and a second channel **310-2** has second connectors **116** at a second position between the ends of second channel **310-2**, wherein the position of the second connectors **116** in second channel **310-2** is staggered relative to the position of the first connectors **116** in first channel **310-1**. Each of connectors **116** may be capacitors or resistors.

(31) Connectors **116** corresponding to first channel **310-1** may be positioned at either end such that positive trace **112** and negative trace **114** may each be a continuous trace over their respective lengths $L_{sub.112}$ and $L_{sub.114}$.

(32) Staggered Position of Connectors in Other Channels

(33) Second channel **310-2** may be formed with first portion **112-1** of positive trace **112** extending a distance $D_{sub.112-1}$, second portion **112-2** extending a distance $D_{sub.112-2}$, and crossover connector **116** positioned between first portion **112-1** and second portion **112-2** such that positive trace **112** is discontinuous. Connector **116** corresponding to positive trace **112** of second channel **310-2** may be positioned between first portion **112-1** and second portion **112-2** a distance $D_{sub.112-1}$ along the length $L_{sub.112}$ of positive trace **112** not near either end of second channel **310-2**.

(34) Second channel **310-2** may be formed with first portion **114-1** of negative trace **114** extending a distance $D_{sub.114-1}$, second portion **114-2** extending a distance $D_{sub.114-2}$, and crossover connector **116** positioned between first portion **114-1** and second portion **114-2** such that negative trace **14** is discontinuous. Connector **116** corresponding to negative trace **114** of second channel **310-2** may be positioned between first portion **114-1** and second portion **114-2** a distance $D_{sub.114-1}$ along the length $L_{sub.114}$ of negative trace **114** not near either end of second channel **310-2**.

(35) Crossover Connections

(36) Still referring to channel **310-2** of FIG. 3, first portion **112-1** of positive trace **112** may be configured on a first side of first portion **114-1** of negative trace **114** and second portion **112-2** of positive trace **112** may be positioned on a second side of second portion **114-2** of negative trace **114**. Thus, first portion **112-1** of positive trace **112** of second channel **310-2** may be positioned adjacent to negative trace **114** of first channel **310-1**. However, second portion **112-2** of positive

trace **112** of second channel **310-2** is positioned away from negative trace **114** of first channel **310-1**, reducing the likelihood of crosstalk between positive trace **112** of second channel **310-2** and negative trace **114** of first channel **310-1**.

(37) In some embodiments, third channel **310-3** may be formed similar to first channel **310-1** because of the configuration of second channel **310-2** and fourth channel **310-4** may be formed similar to second channel **310-2** because of the configuration of third channel **310-3**. In other embodiments (not shown), one or more of third channel **310-3** and fourth channel **310-4** may be formed with connectors **116** positioned staggered relative to adjacent channels **310** at any position along the length of the channel **310** between either end.

(38) FIG. 4 depicts a common implementation of channels **410** for communicating signals in information handling system **100**. As depicted in FIG. 4, each channel **410** is configured with positive trace **412** with connector **422** comprising posts **422A** and **422B**, wherein post **422A** may receive signals from a source and transmit the signals to post **422B** for transmission along positive trace **412**. Furthermore, each channel **410** is configured with negative trace **414** with connector **424** comprising posts **424A** and **424B**, wherein post **424A** may receive signals from the source and transmit the signals to post **424B** for transmission along negative trace **414**. As power and processing speeds increase, crosstalk is likely to occur between negative trace **414** of channel **410-1** and positive trace **412** of channel **410-2**, between negative trace **414** of channel **410-2** and positive trace **412** of channel **410-3** or negative trace **414** of channel **410-3** and positive trace **412** of channel **410-4**. Connectors **422** and **424** may comprise capacitors or resistors.

(39) FIG. 5 depicts an embodiment of a system for preventing crosstalk between adjacent channels **510** in an information handling system **100**. As depicted in FIG. 5, each connector **522** of first channel **510-1** and third channel **510-3** may be configured with connector posts **522A**, **522B** positioned at an end of positive trace **512** such that positive trace **512** is continuous over length L_{512} . Each connector **524** of first channel **510-1** and third channel **510-3** may also be configured with connector posts **524A**, **524B** positioned at an end of negative trace **514** such that negative trace **514** is continuous over length L_{514} .

(40) Each of second channel **510-2** and fourth channel **510-4** may be configured with positive trace **512** comprising first portion **512-1**, second portion **512-2** and connector **522** comprising posts **522A**, **522B** positioned between first portion **512-1** and second portion **512-2** a distance D_{512-1} from the end of positive trace **512** such that a second portion **512-2** of positive trace **512** is located on an opposite side of negative trace **514** and positive trace **512** is discontinuous over length L_{512} . Each of second channel **510-2** and fourth channel **510-4** may also be configured with negative trace **514** comprising first portion **514-1**, second portion **514-2** and connector **524** comprising posts **524A**, **524B** positioned a distance D_{514-1} from the end of negative trace **514** such that negative trace **514** is discontinuous over length L_{514} .

(41) Crossover Connection Based on Breakout Pattern

(42) Still referring to FIG. 5, first portion **512-1** of positive trace **512** may be configured on a first side of first portion **514-1** of negative trace **514** and second portion **512-2** of positive trace **512** may be positioned on a second side of negative portion **514-2**. Thus, first portion **512-1** of positive trace **512** of second channel **510-2** may be positioned adjacent to negative trace **514** of first channel **510-1**. However, second portion **512-2** of positive trace **512** of second channel **510-2** is positioned away from negative trace **514** of first channel **510-1**, reducing the likelihood of crosstalk between positive trace **512** of second channel **510-2** and negative trace **514** of first channel **510-1**. As depicted in FIG. 5, both posts **522A**, **522B** of connector **522** of positive trace **512** are located in-line relative to positive trace **512** and both posts **524A**, **524B** of connector **524** of negative trace **514** are located in-line relative to negative trace **514**. In these embodiments, crossover may occur in a breakout pattern comprising positive trace crossover **526** and negative trace crossover **528**. Connectors **522** and **524** may comprise capacitors or resistors.

(43) Dual Crossover Connector Package

(44) In some embodiments, crossover occurs at a capacitor crossover connector **650**. Referring to FIG. **6**, in some embodiments, second channel **610-2** may be configured with capacitive crossover connector **650** such that first post **622A** of connector **622** is located in line with first portion **612-1** of positive trace **612**, second post **622B** is located in-line with second portion **612-2** of positive trace **612**, first post **624A** of connector **624** is located in line with first portion **614-1** of negative trace **614**, and second post **624B** is located in-line with second portion **614-2** of negative trace **614**. In some embodiments, crossover connector **650** comprises a capacitor.

(45) Referring to FIG. **7**, in some embodiments, crossover occurs at a resistor crossover connector **760**. Referring to FIG. **7**, in some embodiments, first channel **710-1** and third channel **710-3** do not have a connector along traces **712** or **714**. Each of second channel **710-2** and **710-4** may be configured with zero ohm resistor crossover connector **760** such that first positive trace post **722A** of connector **722** is located in line with first portion **712-1** of positive trace **712**, second positive trace post **722B** of connector **722** is located in-line with second portion **612-2** of positive trace **612**, first negative trace post **624A** of connector **724** is located in line with first portion **614-1** of negative trace **614**, and second negative trace post **624B** of connector **724** is located in-line with second portion **614-2** of negative trace **614**.

(46) Referring to FIG. **8**, a crossover connector **800** comprises a plurality of layers **52**. FIG. **8** depicts six layers **52**, but the invention is not limited such that the number of layers **52** may be based on providing a desired capacitance or resistance. As depicted in FIG. **8**, each layer **52** comprises base **54** with two positive pads **56** and two negative pads **60**. Base **54** may comprise nonconducting material, for example ceramic.

(47) Some levels **52** comprise filament **58** connecting either the two positive pads **56** or the two negative pads **60**.

(48) Each of the Pads are Flush With or Recessed From the Sides of the Base

(49) Referring to layer **52-1**, at least one layer **52** comprises positive pad **56-1** configured with a first side (e.g., side **62-1**) approximately flush relative to first edge **64-1** of base **54** and a second side (e.g. side **62-2**) recessed from fourth edge **64-4** of base **54**. Layer **52-1** further comprises positive pad **56-2** configured with a first side (e.g., side **68-1**) approximately flush relative to second edge **64-2** of base **54** and a second side (e.g. side **68-2**) recessed from third edge **64-3** of base **54**. Layer **52-1** further comprises negative pad **60-1** configured with a first side (e.g., side **66-1**) approximately flush relative to first edge **64-1** of base **54** and a second side (e.g. side **66-2**) recessed from second edge **64-2** of base **54**. Layer **52-1** further comprises negative pad **60-2** configured with a first side (e.g., side **70-1**) recessed from third edge **64-3** of base **54** and second side (e.g. side **70-2**) approximately flush relative to fourth edge **64-4** of base **54**.

(50) Thus, crossover connector **800** comprises a plurality of layers **52**, wherein each layer **52** in crossover connector **800** comprises base **54** and two positive pads **56** and two negative pads **60**, wherein at least one layer has a filament **58** between two positive pads **56**, at least one layer **52** has a filament **58** between two negative pads **60**, and at least one layer **52** has no filament between the two positive pads **56** or the two negative pads **60**. Crossover connector **800** may comprise resistors or capacitors.

(51) A first set (e.g., layer **52-1**) of layers **52** comprise positive pads **56-1** having a first side **62-1** flush relative to a first edge **64-1** of base **54**, a second side **62-2** recessed relative to a fourth edge **64-4** of base **54**, positive pads **56-2** having a first side **68-1** flush relative to second edge **64-2** of base **54** and a second side **68-2** recessed relative to third edge **64-3** of base **54** and filament **58** connecting positive pads **56-1** and **56-2**. The first set of layers **52** further comprise negative pads **60-1** having a first side **66-1** flush relative to first edge **64-1** of base **54**, a second side **66-2** recessed relative to second edge **64-2** of base **54** and negative pads **60-2** having a first side **70-1** recessed relative to third edge **64-3** of base **54** and a second side **70-2** flush relative to fourth edge **64-4** of base **54**.

(52) A second set (e.g., layer **52-6**) of layers **52** comprise positive pads **56-1** having a first side **62-1**

recessed relative to first edge **64-1** of base **54**, a second side **62-2** flush relative to a fourth edge **64-4** of base **54**, positive pads **56-2** having a first side **68-1** recessed relative to second edge **64-2** of base **54** and a second side **68-2** flush relative to third edge **64-3** of base **54**. The second set of layers **52** further comprise negative pads **60-1** having a first side **66-1** recessed relative to first edge **64-1** of base **54**, a second side **66-2** flush relative to second edge **64-2** of base **54** and negative pads **60-2** having a first side **70-1** flush relative to third edge **64-3** of base **54** and a fourth side **70-2** recessed relative to fourth edge **64-4** of base **54** and filament **58** connecting negative pads **60-1** and **60-2**.

(53) A third set (e.g., layer **52-3**) of layers **52** comprise positive pads **56-1** having a first side **62-1** flush relative to a first edge **64-1** of base **54**, a second side **62-2** recessed relative to a fourth edge **64-4** of base **54**, positive pads **56-2** having a first side **68-1** flush relative to second edge **64-2** of base **54** and a second side **68-2** recessed relative to third edge **64-3** of base **54**. The third set of layers **52** further comprise negative pads **60-1** having a first side **66-1** flush relative to first edge **64-1** of base **54**, a second side **66-2** recessed relative to a second edge **64-2** of base **54** and negative pads **60-2** having a first side **70-1** recessed relative to second edge **64-2** of base **54** and a second side **70-2** flush relative to fourth edge **64-4** of base **54**. The third set of layers may not have filament **58** between positive pads **56** or negative pads **60**.

(54) A fourth set (e.g., layer **52-4**) of layers **52** comprise positive pads **56-1** having a first side **62-1** recessed relative to first edge **64-1** of base **54**, a second side **62-2** flush relative to a fourth edge **64-4** of base **54** and positive pads **56-2** having a first side **68-1** recessed relative to second edge **64-2** of base **54** and a second side **68-2** flush relative to third edge **64-3** of base **54**. The fourth set of layers **52** further comprise negative pads **60-1** having a first side **66-1** recessed relative to first edge **64-1** of base **54**, a second side **66-2** flush relative to a second edge **64-2** of base **54** and negative pads **60-2** having a first side **70-1** flush relative to third edge **64-3** of base **54** and a second side **70-2** recessed relative to fourth edge **64-4** of base **54**. The fourth set of layers may not have filament **58** between positive pads **56** or negative pads **60**.

(55) FIGS. **9A** and **9B** depict an exploded perspective view and an assembled perspective view, respectively, of a dual crossover connector package **900** for use as a crossover connection such as crossover connection **650** or **760** as depicted in FIG. **6** or **7**. FIGS. **9A** and **9B** depict a dual crossover connector package **900** comprising six layers **52**. As visible in FIG. **9A**, pads **56** and **60** may be positioned on layers **54** such that at least one side of each positive pad **56** is flush with a side of base **54** and at least one side is recessed relative to base **54** and at least one side of each negative pad **60** is flush with a side of base **54** and at least one side is recessed relative to base **54**.

(56) Connection Between Layers

(57) Two or more positive pads **56** or negative pads **60** on multiple layers **52** may be connected. Referring to FIG. **10**, a close-up partial perspective view of one embodiment of a dual crossover connector package **800** illustrates a first offset (O.sub.1) between positive pads **56** relative to a first edge of base **54** and a second offset (O.sub.2) between positive pads **56** relative to a second edge of base **54**. In this configuration, a coupling on a first side may connect positive pads **56** corresponding to layers **52-1**, **52-3** and **52-5** and a coupling on a second side may connect layers **52-2**, **52-4** and **52-6** into a set of layers **52**. As more positive pads **56** and negative pads **60** are connected, capacitance may increase, which may be more advantageous for preventing crosstalk between adjacent channels.

(58) Referring to FIG. **11**, a close-up perspective view of one embodiment of a dual crossover connector package **1100** illustrates couplings that connect positive pads **56** or negative pads **60** of multiple layers **52**. As depicted in FIG. **11**, bases **54** are not fully depicted for clarity.

(59) As depicted in FIG. **11**, contacts **80-1** and **80-2** are coupled to positive pads **56** and contacts **82-1** and **82-2** are coupled to negative pads **60**.

(60) Contact **80-1** may be coupled to positive pads **56-1** on selected layers **52-2**, **52-4** and **52-6** of a plurality of layers **52-1** to **52-6**. Although not visible in FIG. **11**, contact **80-2** may be coupled to positive pads **56-2** on selected layers **52-2**, **52-4** and **52-6**. Connector **86-1** may be coupled to

positive pads **56** on layers **52-1**, **52-3** and **52-5** of the plurality of layers **52-1** to **52-6**. Connector **86-3** may be coupled to positive pads **56-2** on layers **52-1**, **52-3** and **52-5** of the plurality of layers **52-1** to **52-6**.

(61) Although not visible in FIG. **11**, contact **82-1** may be coupled to negative pads **60-1** on selected layers **52-2**, **52-4** and **52-6** of a plurality of layers **52-1** to **52-6**. Contact **82-2** may be coupled to negative pads **60-2** on selected layers **52-2**, **52-4** and **52-6** of the plurality of layers **52-1** to **52-6**. Connector **86-2** may be coupled to negative pads **60-2** on layers **52-2**, **52-4** and **52-6** of the plurality of layers **52-1** to **52-6**.

(62) Referring to FIG. **12**, manufacturing an information handling system **100** with a system **1200** for preventing crosstalk may comprise identifying a channel **1210** in which crosstalk is possible and positioning a system for preventing crosstalk in the channel **1210**.

(63) As depicted in FIG. **12**, channel **1210-1** connects processor **12** and component **1208** on card **1206-1**. Card **1206-1** may be installed in PCI slot **1204**. As depicted in FIG. **12**, the geometric center **1212** of channel **1210-1** is at a first location on motherboard **202**. However, a system for preventing crosstalk may be more effective if a crossover connector such as connector **650** or **750** is located at point **1214-1**, which is closer to processor **12**.

(64) Also depicted in FIG. **12**, channel **1210-2** connects processor **12** to hard drive **1216** using cable **1218** between connector **1220** and hard drive **1216**. Cable **1218** may have very low losses such that between processor **12** and hard drive **1216**, only a channel portion **1210-2** may have losses. As depicted in FIG. **12**, geometric center **1212-1** may be the actual geometric center between processor of processor **12** and hard drive **1216**, but geometric center **1212-2** may be the geometric center of channel **1210-2** associated with losses. Accordingly, a system for preventing crosstalk may be located at point **1214-2** on motherboard **202** between processor **12** and cable connector **1220**.

(65) Also depicted in FIG. **12**, channel **1210-3** connects processor **12** and component **1222** on card **1206-2**, which is connected to motherboard **202** using cable **1218** connected to cable connector **1224**. Cable **1218** may have losses such that between processor **12** and component **1222**, geometric center **1212** of channel **1210-3** is located along cable **1218**. Accordingly, a system for preventing crosstalk may be located at point **1214-3** on card **1206-2** closer to component **1222**.

(66) The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments which fall within the true spirit and scope of the disclosure. Thus, to the maximum extent allowed by law, the scope of the disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A system for preventing crosstalk between two adjacent channels, the system comprising: a first channel having a first channel length, the first channel comprising: a first positive trace having a first positive trace length; a first negative trace having a first negative trace length; and a connector positioned at a first location along the first positive trace length, wherein the first positive trace is on a first side of the first negative trace; a second channel having a second channel length, the second channel comprising: a second positive trace comprising a first positive trace portion having a first positive trace portion length and a second positive trace portion having a second positive trace portion length; a second negative trace comprising a first negative trace portion having a first negative trace portion length and a second negative trace portion having a second negative trace portion length; and a crossover connector comprising a positive trace crossover and a negative trace crossover, the crossover connector is located on a card and positioned a distance from a source, wherein: the first positive trace portion of the second positive trace is on the first side of the first

negative trace portion of the second negative trace and is coupled to a first positive trace post of the crossover connector, wherein the first negative trace portion of the second negative trace is coupled to a first negative trace post of the crossover connector; the second positive trace portion of the second positive trace is on the second side of the second negative trace portion of the second negative trace and is coupled to a second positive trace post in the crossover connector, wherein the second negative trace portion of the second negative trace is coupled to a second negative trace post, the crossover connector further comprising: a plurality of layers, wherein each layer comprises two positive post pads and two negative post pads, wherein at least one layer comprises a filament between the two positive pads; at least one layer comprises a filament between the two negative pads; and at least one layer does not have a filament between the two positive pads or the two negative pads.

2. The system of claim 1, wherein the crossover connector comprises a resistor.

3. The system of claim 1, wherein the crossover connector comprises a capacitor.
